

SEVERITY

1

2

3

4



US ACCIDENTS DATA ANALYSIS

Can we predict accident severity using road,
weather, time and location conditions?

DATA 230 · Mid-Project Presentation

Department of Applied Data Science

San Jose State University

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Fall 2026

AGENDA

- **Dataset introduction** (Saida)
- Data cleaning (Jue)
- Exploratory data analysis (Nino)
- Preliminary ML (Tzu-Yang)
- Tableau dashboard (Jue)

Dataset introduction & motivation

US Accidents (Kaggle): traffic accidents in 49 U.S. states, January 2016 to March 2023

Central question: Can we predict accident severity using road, weather, time and location ?

SIZE

7.7 M

accidents × 46 columns (~3 GB)

FEATURES

When · Where · Weather · Road
features · Day / night

Target: Severity 1–4

WHY THIS DATASET

All four kinds of factors in one
place, at national scale; public, so
the work can be reproduced

Large and messy: hidden duplicates · missing weather values · impossible readings (e.g. 1,087 mph wind) · strong imbalance (about 80% accident under Severity 2)

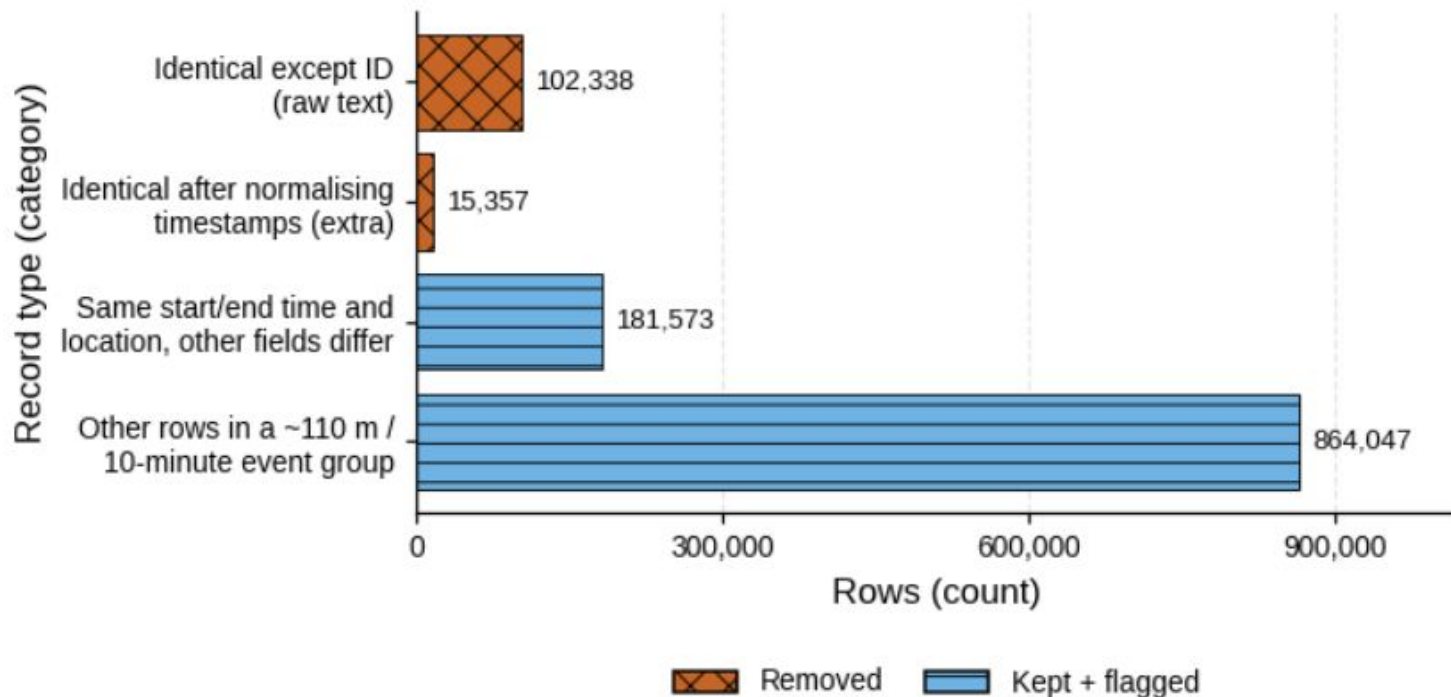
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Preprocessing workflow and rationale

Step	What we did	Why (rationale)
1. Read	Full CSV read in 250,000-row blocks	Use every record; no sampling
2. Remove duplicates	Rows compared without the ID: 117,695 removed	Count each accident once
3. Fix data types	Text → dates, numbers and categories	Correct maths, sorting and charts
4. Invalid values	282 invalid readings → missing, row kept	Remove the error, not the accident
5. Standardize labels	Weather labels harmonised and grouped; wind 24 → 18	Consistent categories for analysis

Duplicates and invalid values

Figure: Duplicates removed vs. repeats kept



117,695

duplicates

282

readings masked:

- 3 physically impossible
- 279 implausible rows

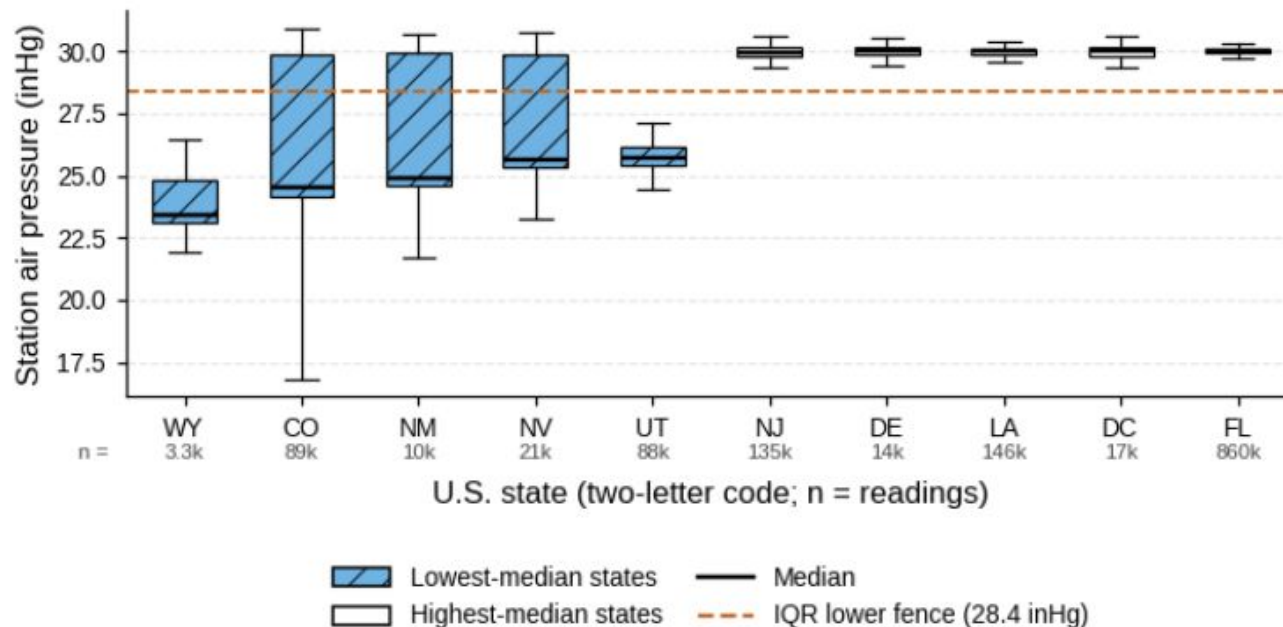
Orange = proven copies, removed. Blue = probable repeats, kept and flagged.

Preprocessing workflow and rationale

Step	What we did	Why (rationale)
1. Read	Full CSV read in 250,000-row blocks, all columns as text	Use every record; no sampling
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5. Standardize labels	Weather labels harmonised and grouped; wind 24 → 18	Consistent categories for analysis
6. Create features	Hour, holiday, rush hour, road type, region	Useful signals for Severity
7. Check outliers	Checked with IQR; reported, not removed	Real extremes, e.g. mountain pressure

Outliers, transformations and merging

Figure: Air pressure by state: low values come from mountain states



Outliers: an IQR rule flags 5.8% of pressure readings, mostly from high-elevation states (WY, CO, NM). Reported, not removed.

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8. Split & leakage	Post-event fields excluded; split by time; repeated reports kept together	Honest test on future accidents
9. Missing values	Kept visible; filled using training rows only	Gaps depend on when data was collected

Missing values are not random

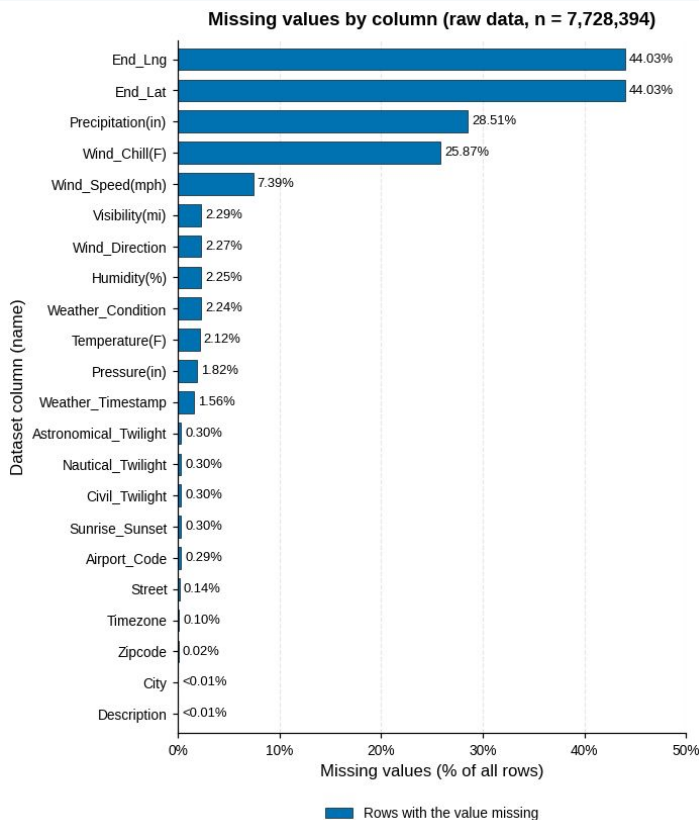
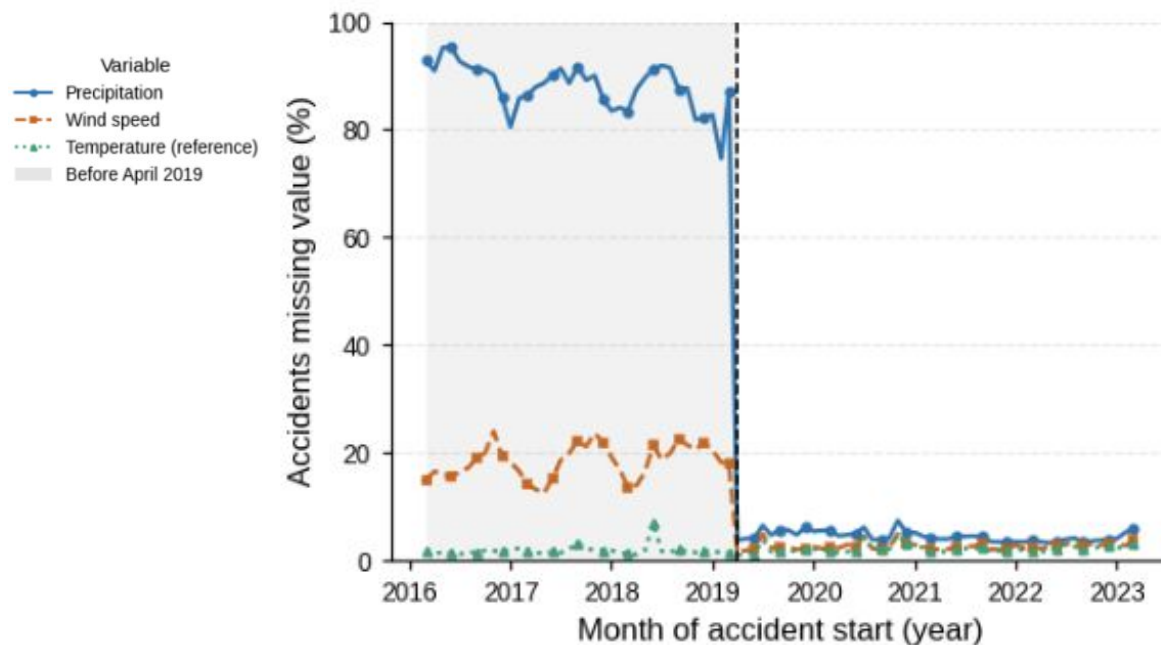


Figure: Missing weather values by month, 2016–2023



87% → 4%: precipitation gaps drop in April 2019, a collection change, not chance.

Decision: keep gaps visible; impute only for modelling, from training rows, by state and month.

Challenges and how we solved them

Challenge	Solution
Memory limits: a 3 GB file crashed a free 12 GB Colab machine	250,000-row blocks and checkpoints: peak memory 6.7 → 3.1 GB
Duplicates hidden by a unique ID	Compared rows without the ID: 117,695 removed
Missing values not random (April 2019)	Kept gaps visible; imputed only from training rows
Data leakage: fields known only after an accident	Time-based split and a role for every column: only 33 safe model inputs

Data cleaning & processing

ROWS

7,610,699

rows after cleaning (from 7,728,394)

DUPLICATES

117,695

duplicate rows removed

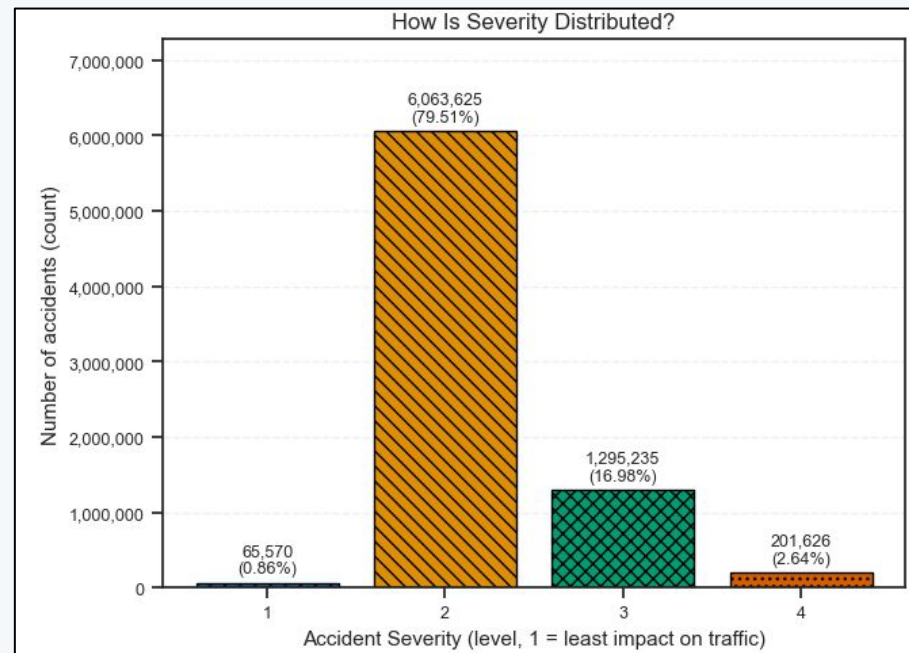
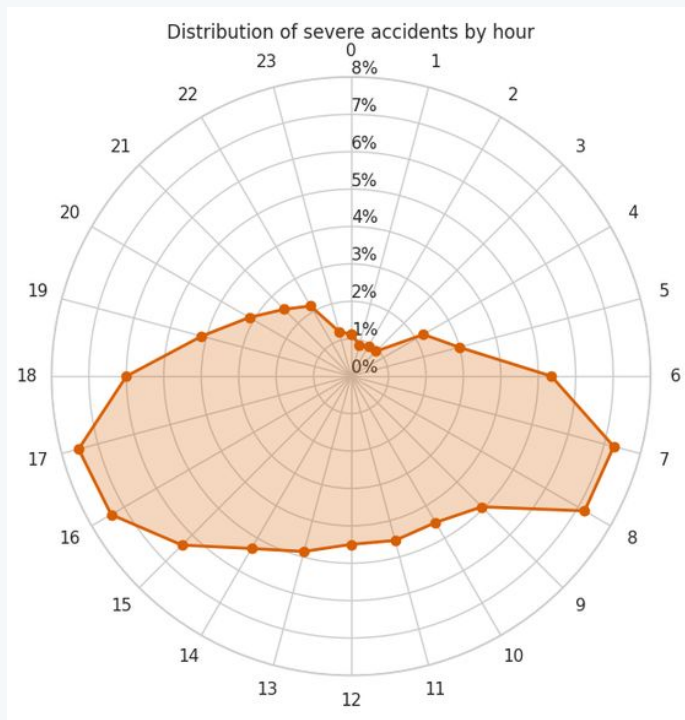
COLUMNS

46 → 79

columns after cleaning; only 33 are
safe model inputs

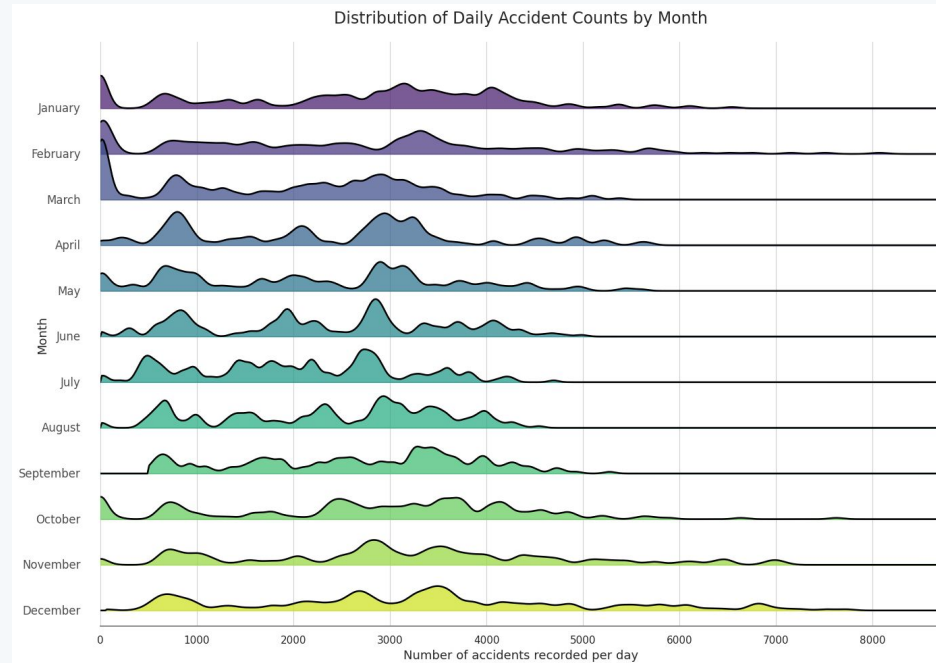
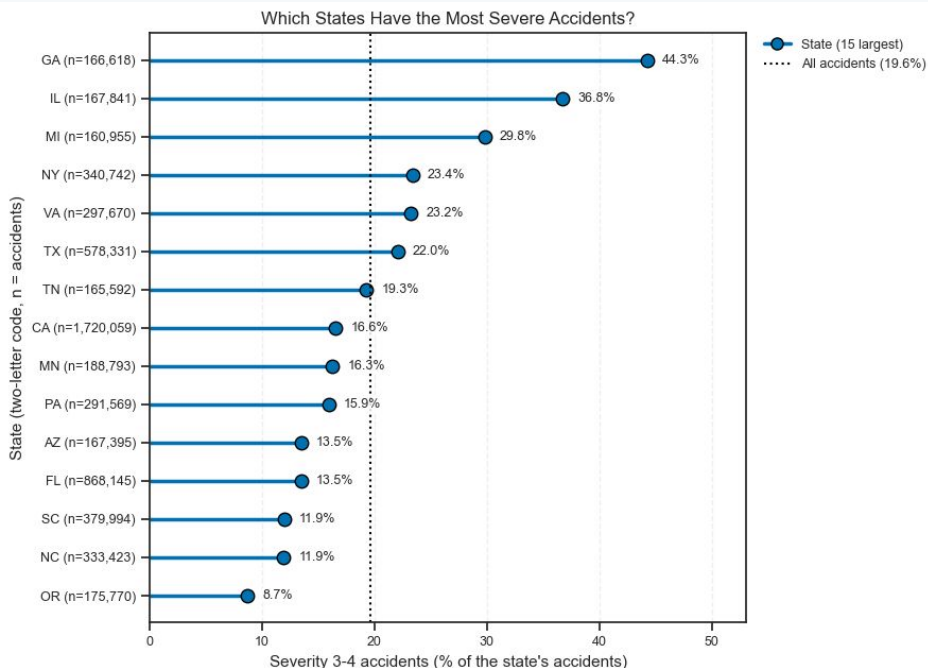
Main goal: fix errors without deleting real accidents.

Preliminary EDA

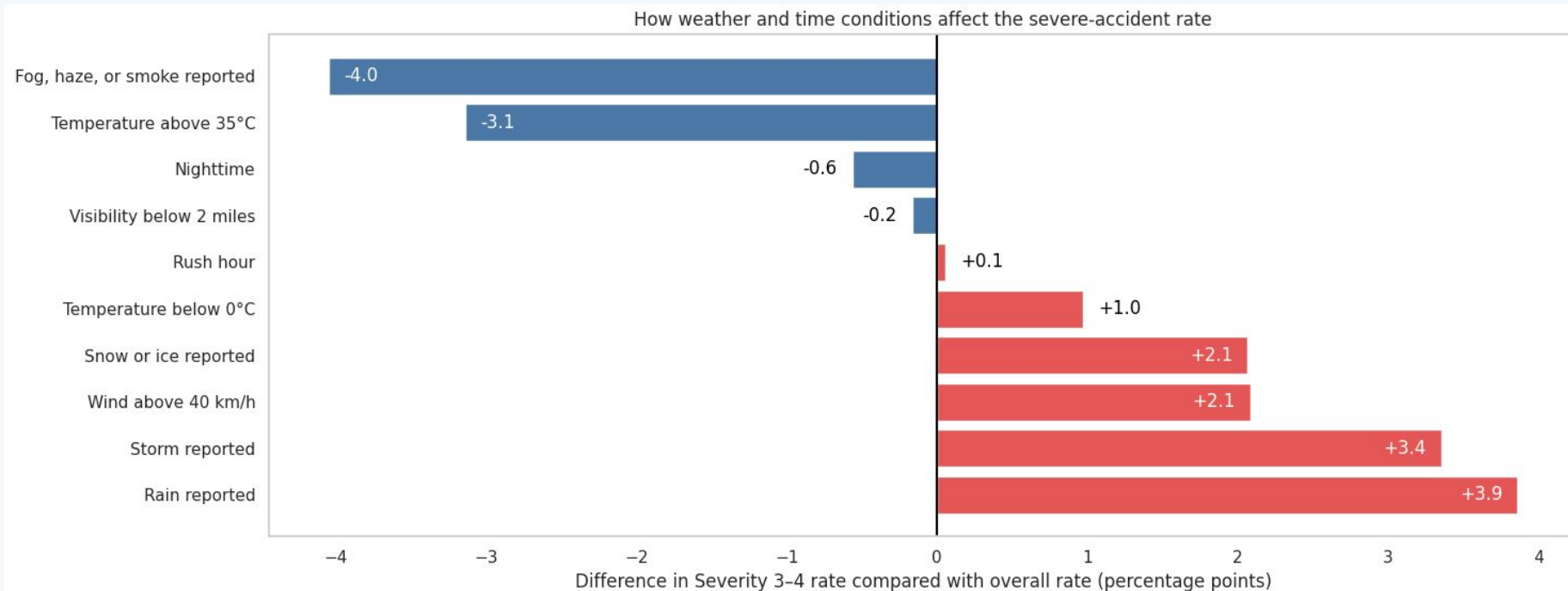


Severity 1 and Severity 4 contain relatively few records, therefore further grouping 1&2 and 3&4 make sense.

Preliminary EDA

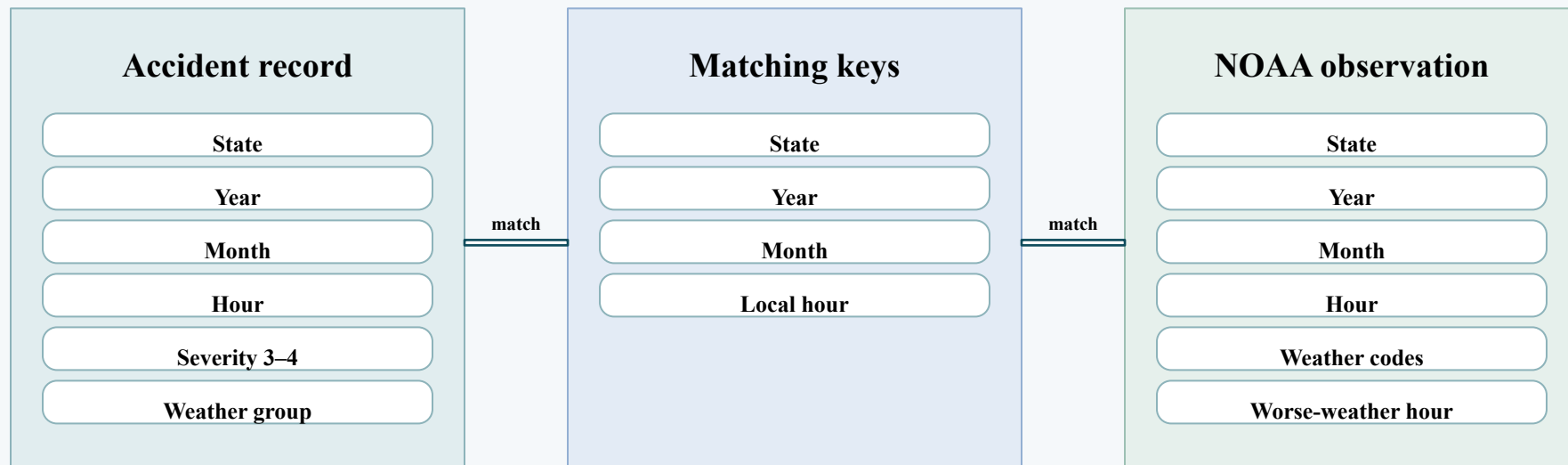


Preliminary EDA



External weather data join

The comparison uses the same state, month, year, and local hour from both datasets



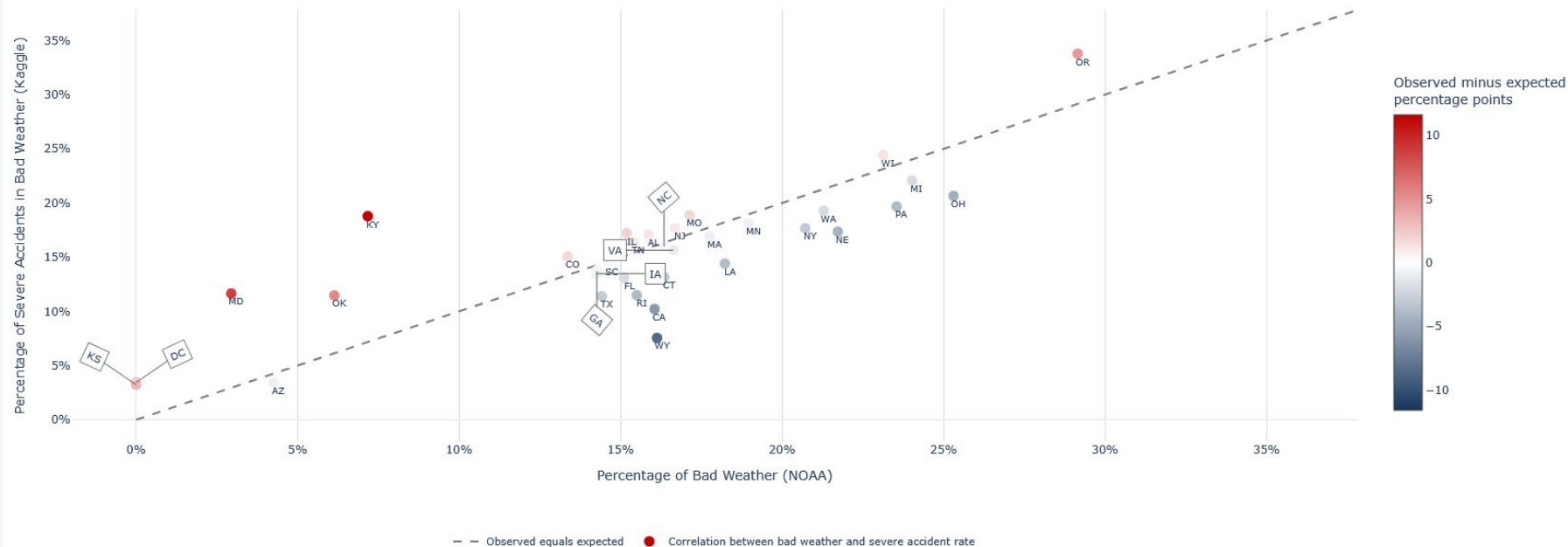
Feature metric:

Observed severe-accident share decreased by expected weather exposure share

Is bad weather statistically more dangerous?

Do severe accidents occur in worse weather more often than expected?

Each point compares severe-accident share with independent NOAA weather exposure for the same state, year, month, and local hour



The chart shows comparison if severe accidents in bad weather in relation of bad weather share, per state

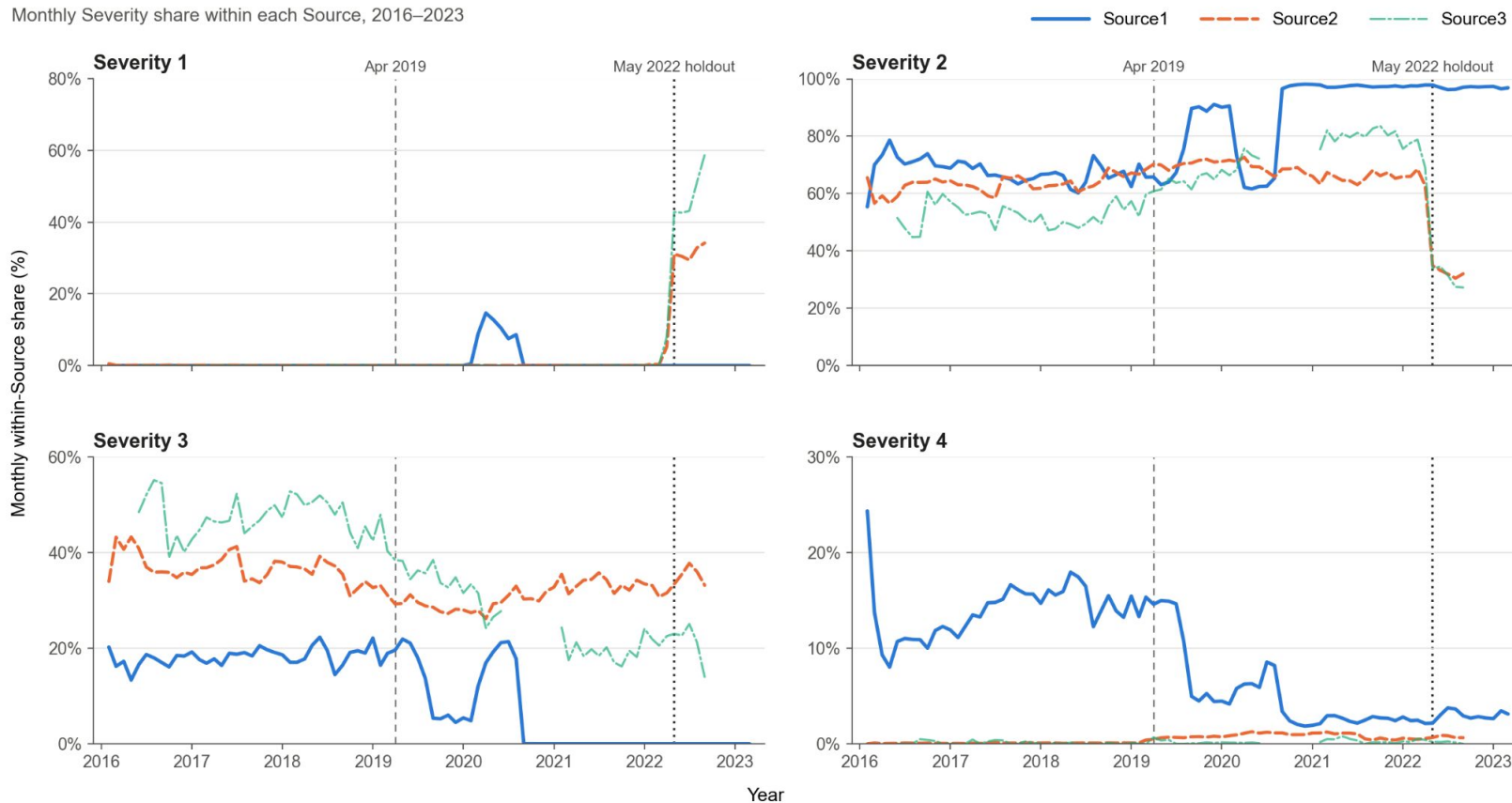
Sources: NOAA Climate Division precipitation data, Global Historical Climatology Network hourly (GHCNh)

Preliminary ML

Target	4-class Severity	Predict Severity 1–4 from road, weather, time, and location information available at accident reporting time.
Features	Predictor selection	Start from 33 candidate features mentioned above. Refine the final feature set based on redundancy, missingness, cardinality, and model-specific diagnostics.
Baseline	Multinomial logistic regression	Train Multinomial Logistic Regression with class weighting , then compare transformations, Random Forest, and Gradient Boosting.
Validation	Time series strategy	Train on accidents before May 1, 2022 and test on May 2022+ ; also compare with an 80/20 grouped random split .
Metrics	Besides accuracy	Primary: Macro F1 . Also report per-class precision/recall, confusion matrix, ordinal MAE, and performance by Source.

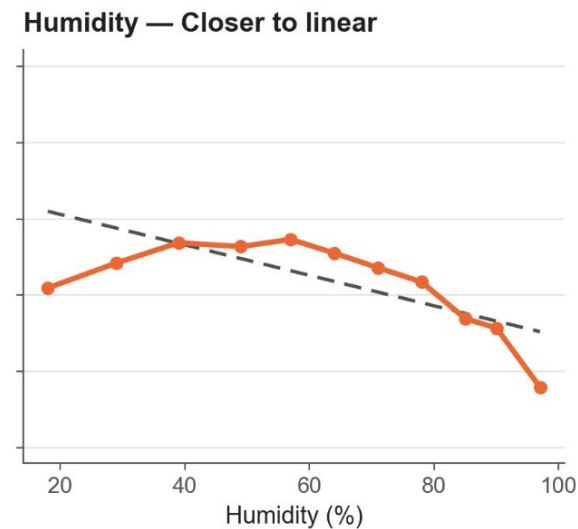
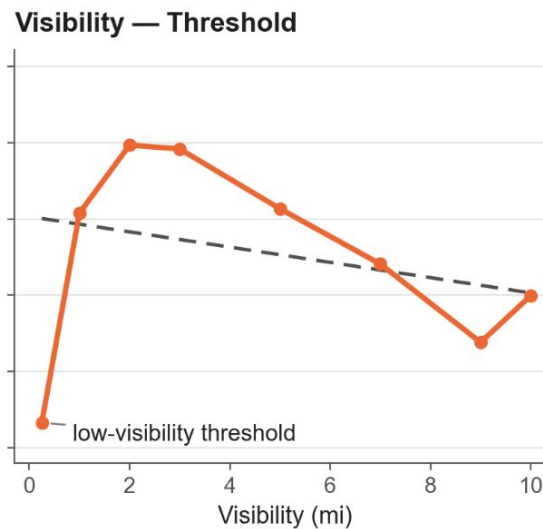
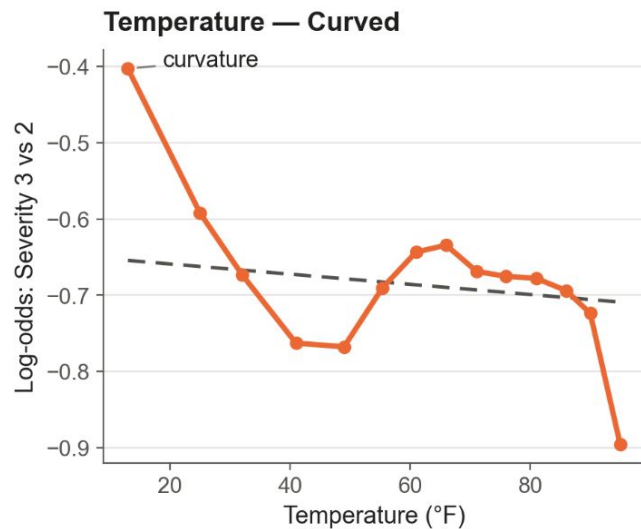
Severity Distributions Shift Differently Across Data Sources

Monthly Severity share within each Source, 2016–2023



One Linear Term Does Not Fit Every Feature

Source2 training data · solid = empirical pattern · dashed = linear reference



Candidate ML features

Time

- Hour, Day of Week, Month, Season
- Is Night, Is Rush Hour

Weather

- Weather Group, Temperature, Humidity
- Visibility, Wind Speed, Precipitation
- Rain Day, Snow or Ice, Low Visibility, Storm
- Extreme Cold, Extreme Hot, High Wind
- Monthly Temperature Anomaly

Road and location

- State
- Junction, Crossing, Traffic Signal, Stop
- Roundabout, Railway, Station, Give Way
- Traffic Feature Count, No Traffic Control

Data quality

Data Quality Score
Missingness flags

Target

Severity (1–4)

Exclude initially

ID, End Time, End Lat, End Lng,
Description

SEVERITY

1

2

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THANK YOU

Questions?

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DATA 230 · San Jose State University · Fall 2026

References

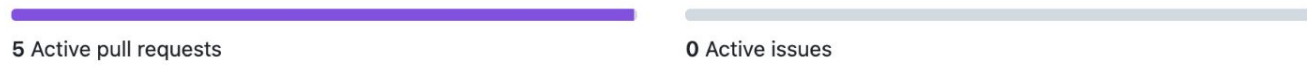
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- Moosavi, S., Samavatian, M. H., Parthasarathy, S., & Ramnath, R. (2019). *A countrywide traffic accident dataset*. arXiv.
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- Pulse
- Contributors
- Community
- Community standards
- Traffic
- Commits
- Code frequency
- Dependency graph
- Network
- Forks
- Actions usage metrics
- Actions performance metrics

September 30, 2026 – October 7, 2026

Period: 1 week ▾

Overview



🔗 5 Merged pull requests	🔗 0 Open pull requests	✅ 0 Closed issues	🕒 0 New issues
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Summary

Excluding merges, **7 authors** have pushed **10 commits** to main and **13 commits** to all branches.

On main, **61 files** have changed and there have been **37,388 additions** and **2 deletions**

Top committers

